PCA Visualisation

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2025-08-08

Preparing the data for PCA

Initial Parameters

```
## [1] "xrf resolution = 200 µm"
```

```
## [1] "analysis will be done on all core"
```

```
## [1] "Feature selection:"
```

```
## Fe, Mn, Ba, S, Ti, Al, Si, K, Ca, Rb, Zr, Sr, P, Ba
```

Load corresponding datasets

```
## [1] "Loading of 200µm data"
```

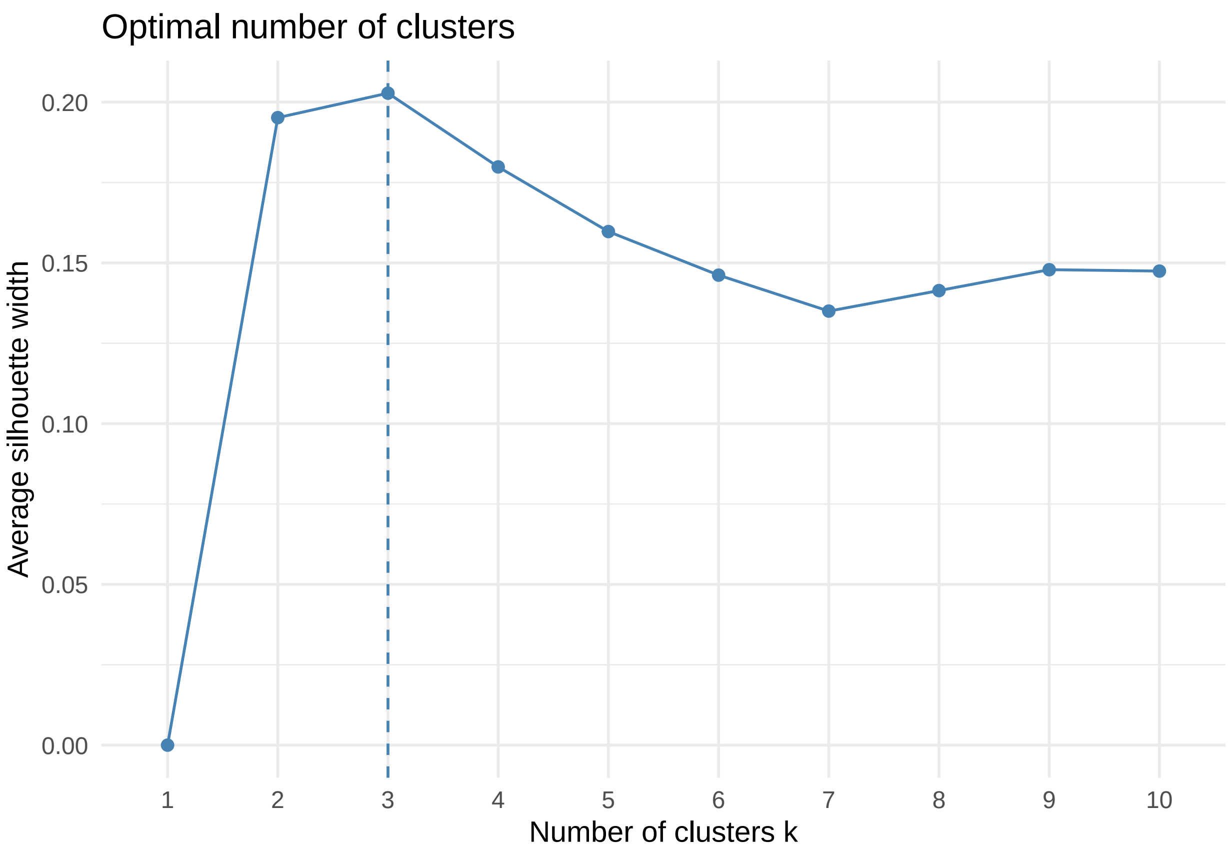
```
## [1] "Corresponding combined dataframe is /Users/maxshore/Documents/Unibe/MasterThesis/masterthesis/R/
```

Prepare transformed data for PCA and clustering

```
df.xrf <- clean_df(df.raw, sel = sel)

if(res_micro == 500){
  df.xrf$raw.200 <- NULL
  df.xrf <- tibble(df.xrf$raw.500)
} else {
  df.xrf$raw.500 <- NULL
  df.xrf <- tibble(df.xrf$raw.200)
} # selection of dataframe
if(l_analysis == 'top_'){
  df.xrf <- df.xrf %>%
    select_top_varves()
} # selection of top varves
df.cs <- closed_sum(df.xrf)
df.clr <- clr_transform(df.xrf)
```

Perform PCA and k-means clustering

```
pca.df <- perform_pca2(df.clr)
X <- scale(df.clr %>% select(-position..mm.))
# Choose optimal number of clusters
f_clust <- fviz_nbclust(X, kmeans, method = "silhouette") +
  theme_minimal()
print(f_clust)
```

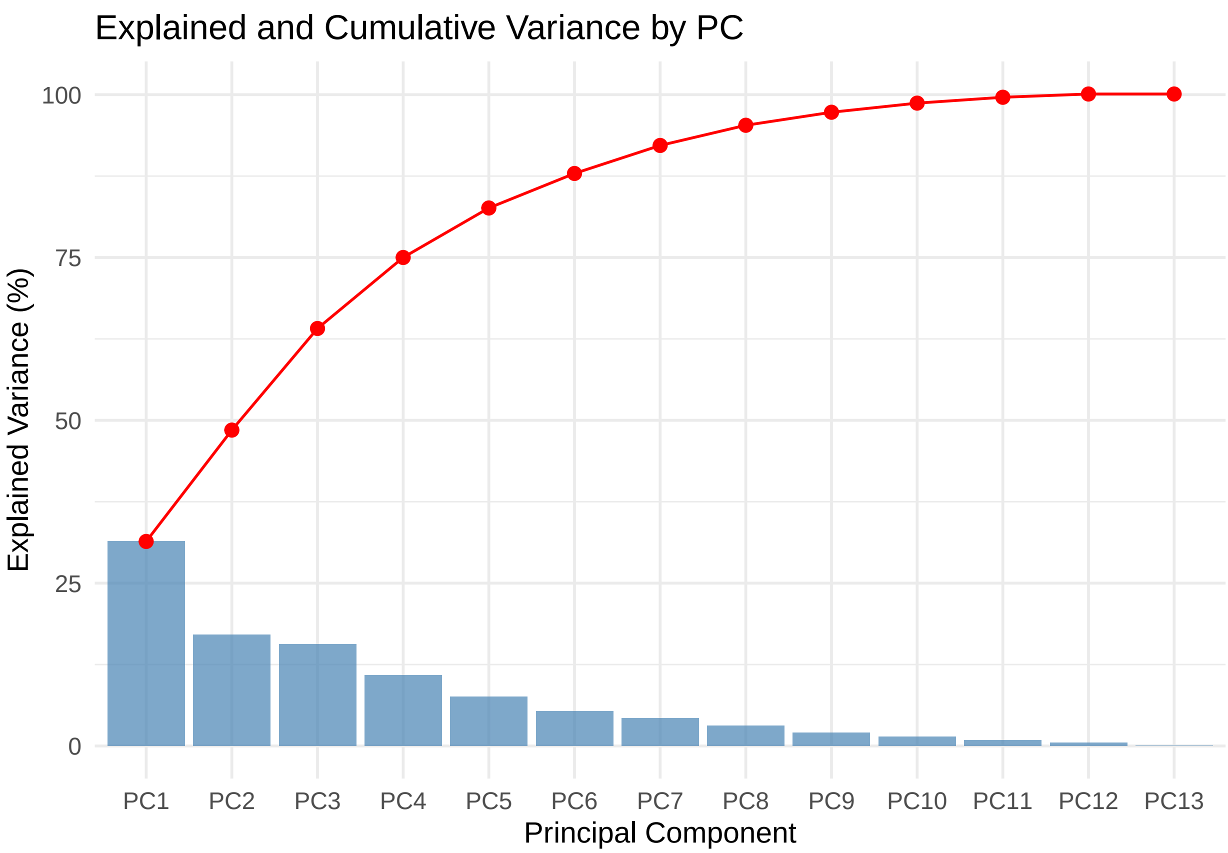



```
n_clust <- f_clust$data
optimal_clust <- as.numeric(n_clust$clusters[which.max(n_clust$y)])


print(paste('Optimal number of clusters is', optimal_clust))
```

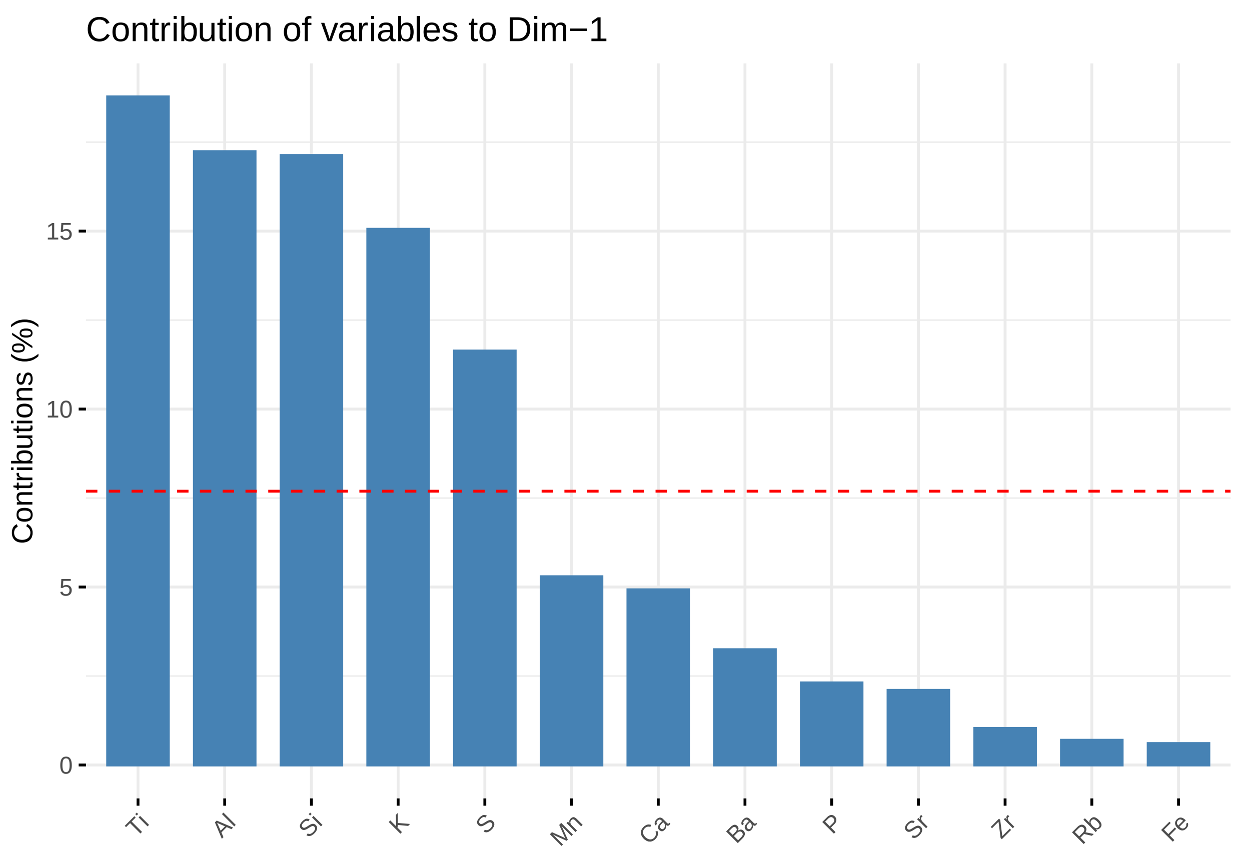
```
## [1] "Optimal number of clusters is 3"
```

```
km <- kmeans(X, centers = optimal_clust, nstart = 25)

scores <- as_tibble(pca.df$x)
loadings <- as_tibble(pca.df$rotation, rownames = "variable")

# Rescale loadings to match sample space
loadings_scaled <- loadings %>%
  mutate(
    PC1 = PC1 * pca.df$sdev[1],
    PC2 = PC2 * pca.df$sdev[2],
    PC3 = PC3 * pca.df$sdev[3]
  )

#  Add metadata for faceting
scores <- scores %>%
  mutate(depth = pca.df$position..mm.) %>%
  mutate(kcluster = as.factor(km$cluster))
```

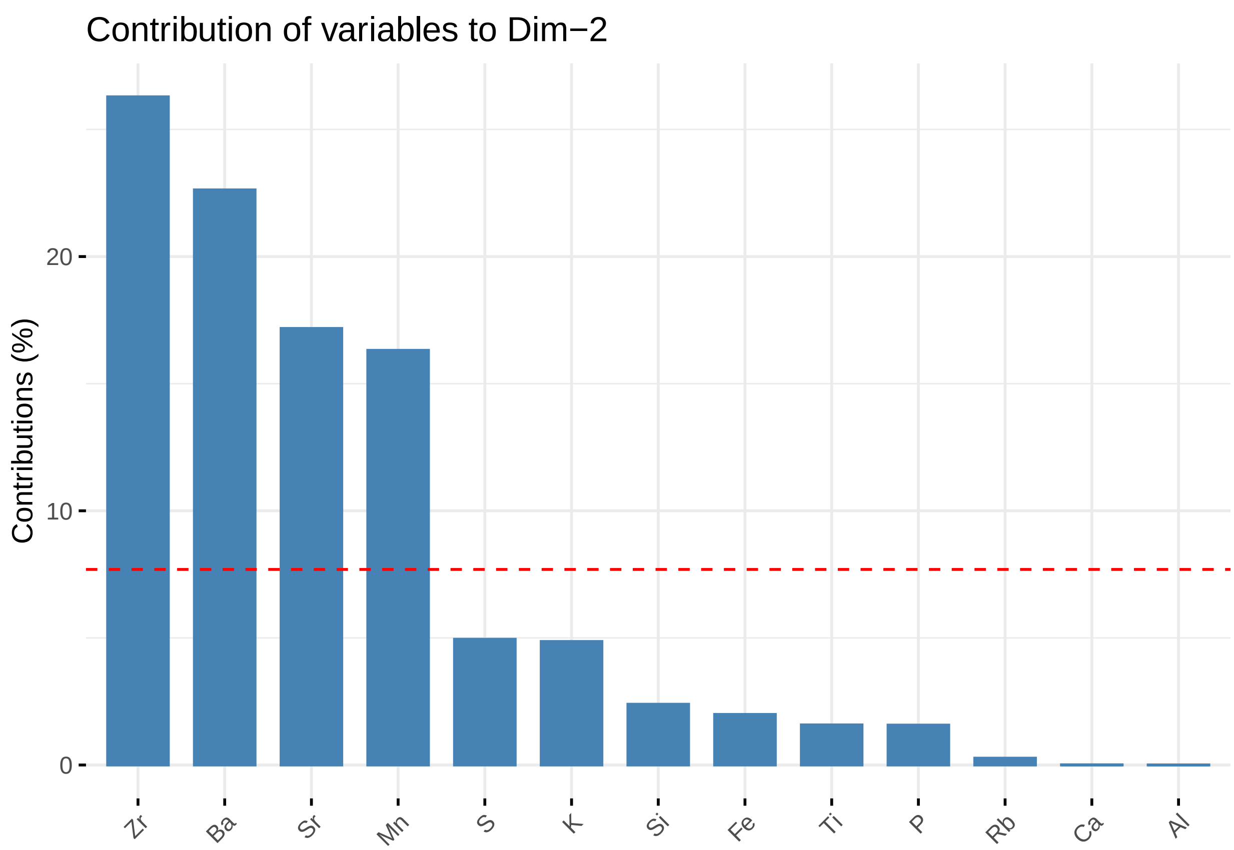
```
var_expl <- round(100 * pca.df$sdev^2 / sum(pca.df$sdev^2), 1)
cum_var <- cumsum(var_expl)
pc_names = pca.df$rotation %>% colnames()
var_pca <- tibble(
  pc = pc_names,
  var_expl = var_expl,
  cum_var = cum_var
)
var_pca <- var_pca %>%
  mutate(pc = factor(pc, levels = pc))

p_var_expl <- ggplot(var_pca, aes(x = pc)) +
  geom_bar(aes(y = var_expl), stat = "identity", fill = "steelblue", alpha = 0.7) +
  geom_line(aes(y = cum_var, group = 1), color = "red", linewidth = 0.5) +
  geom_point(aes(y = cum_var), color = "red", size = 2) +
  labs(
    x = "Principal Component",
    y = "Explained Variance (%)",
    title = "Explained and Cumulative Variance by PC"
  ) +
  theme_minimal()
print(p_var_expl)
```



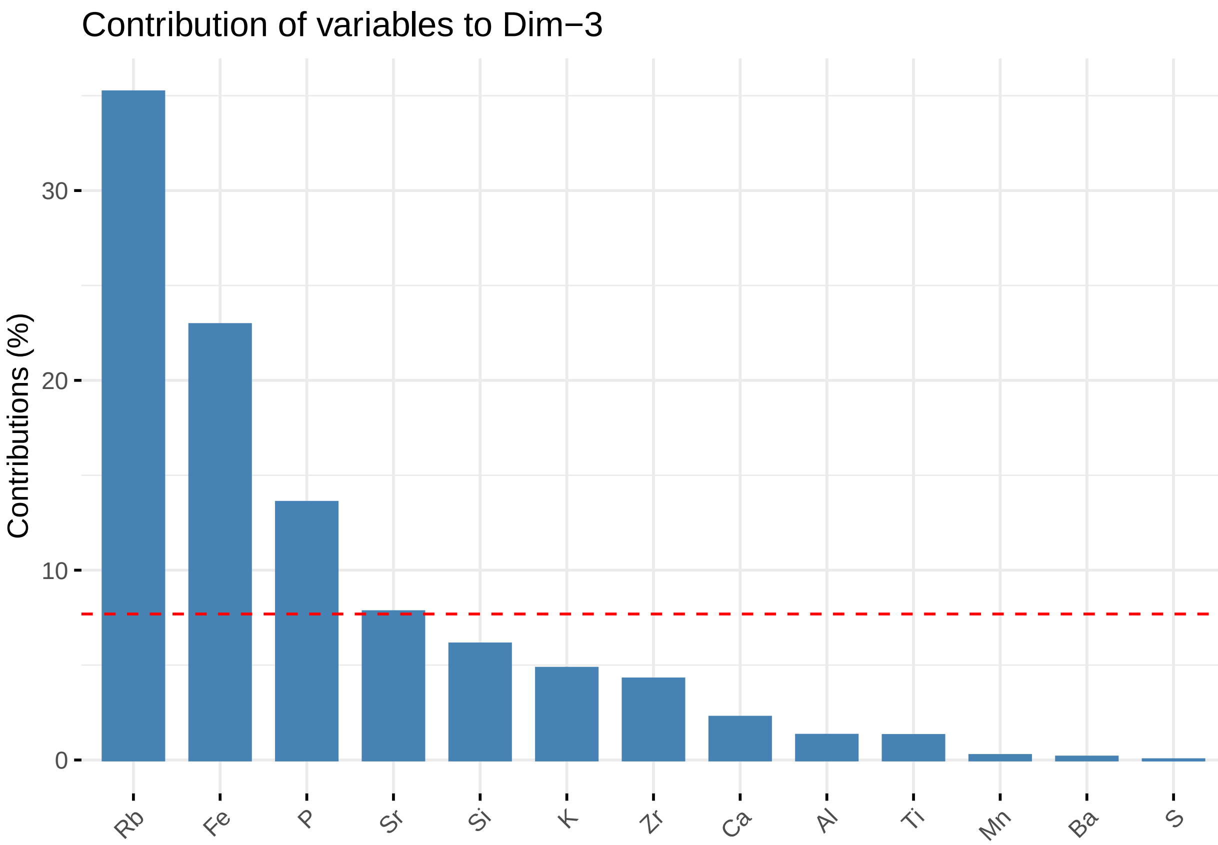

```
fviz_contrib(pca.df, choice='var', axes = c(1))
```




```r
fviz_contrib(pca.df, choice = 'var', axes = c(2))
```

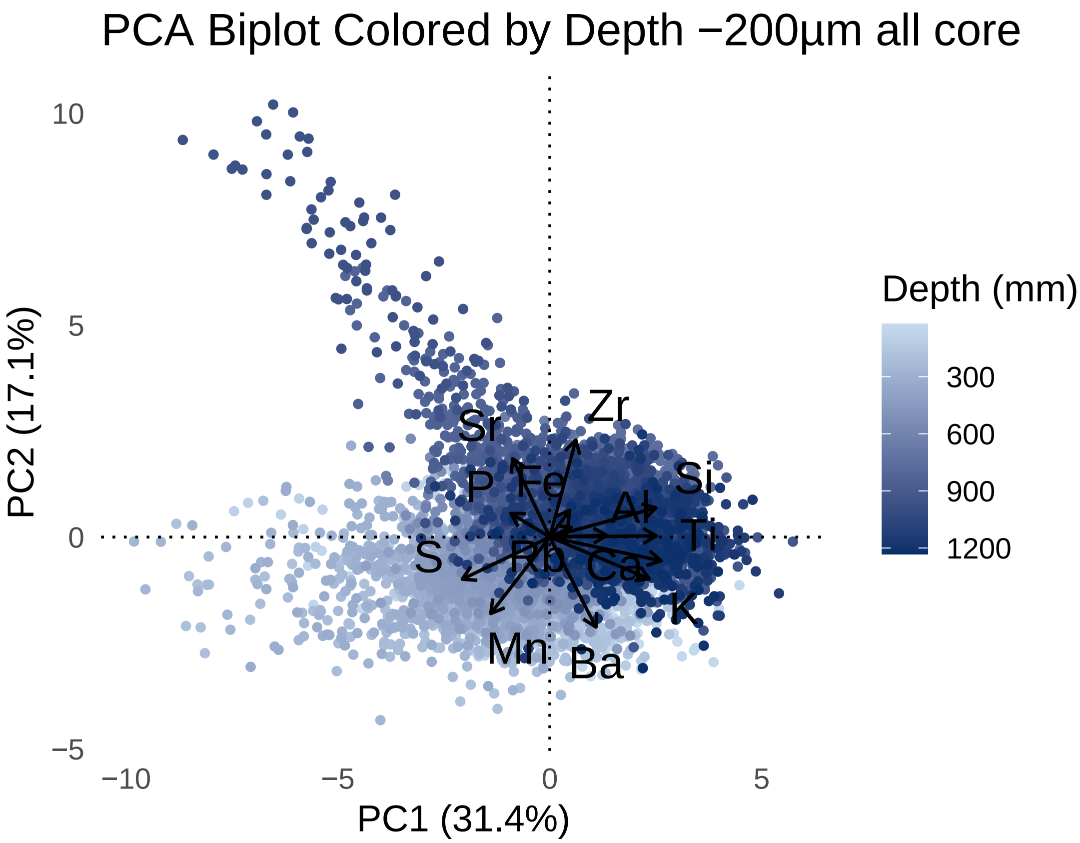



```
fviz_contrib(pca.df, choice='var', axes = c(3))
```

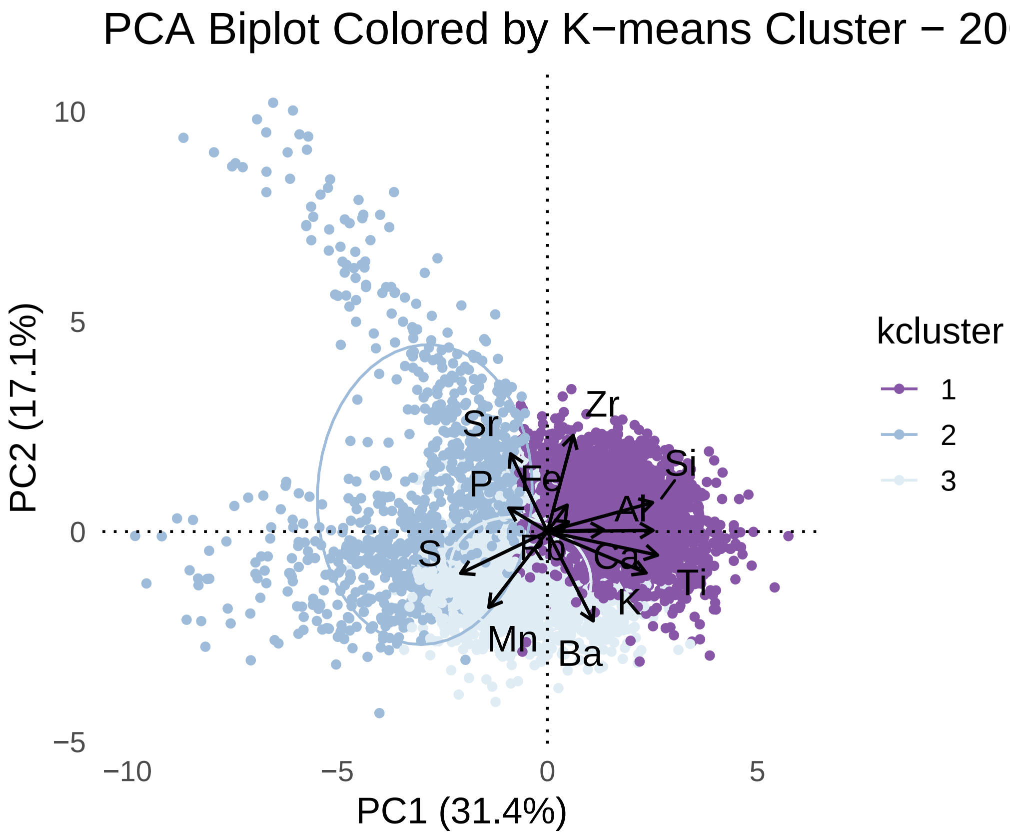

Contribution of variables to Dim−3
Contributions (%)
0
10
20
30
Rb
Fe
P
Sr
Si
K
Zr
Ca
Al
Ti
Mn
Ba
S

Biplots of PC1, PC2 and PC3

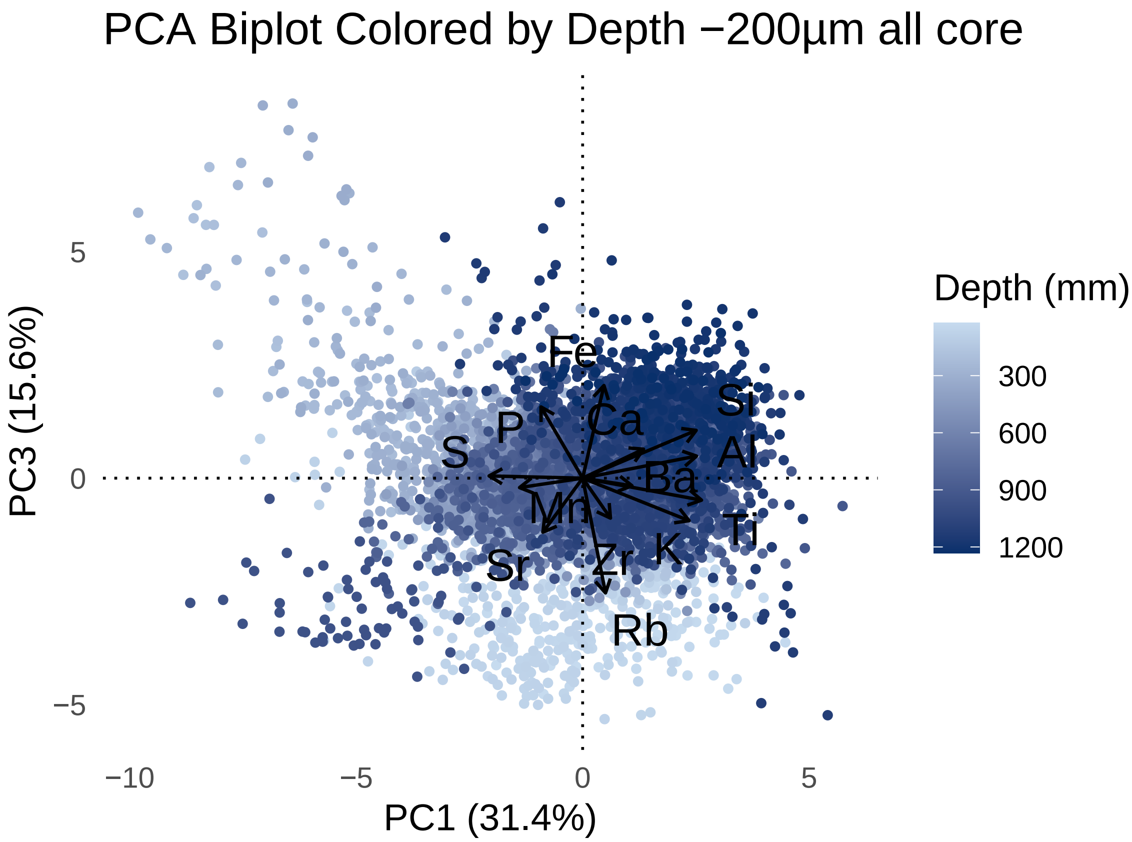
PC1 vs PC2 - Depth facteting



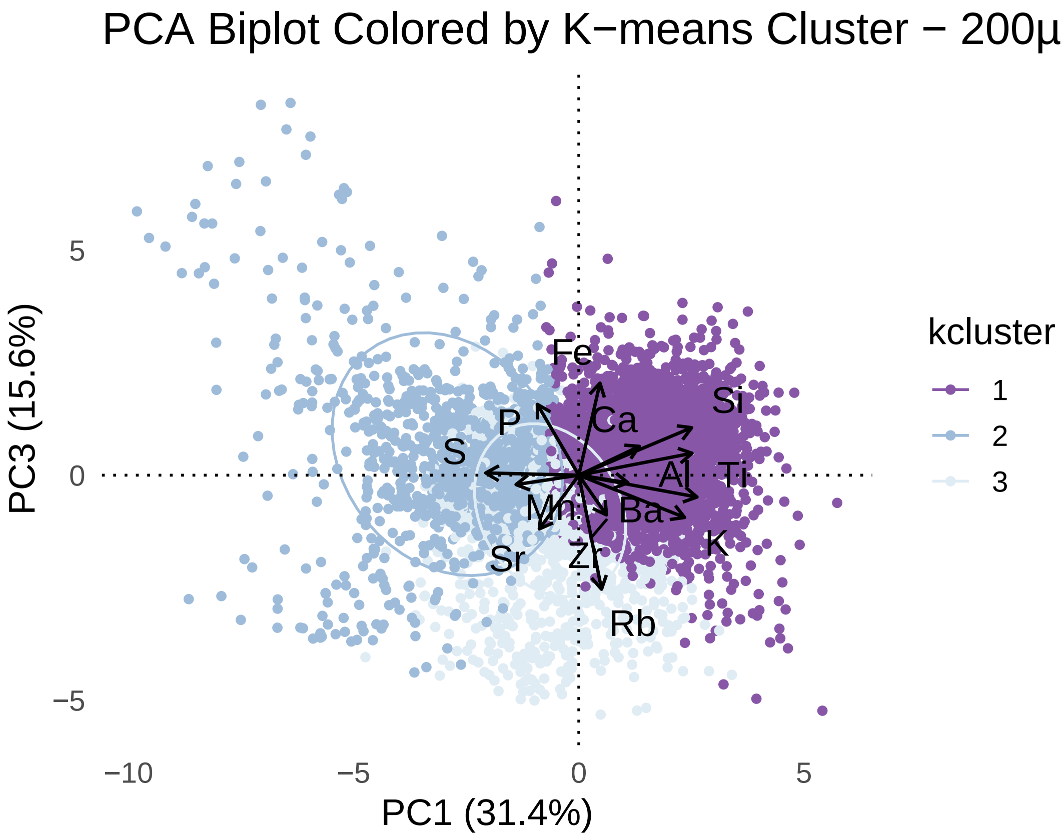
PC1 vs PC2 - K-means cluster faceting



PC1 vs PC3 - Depth faceting

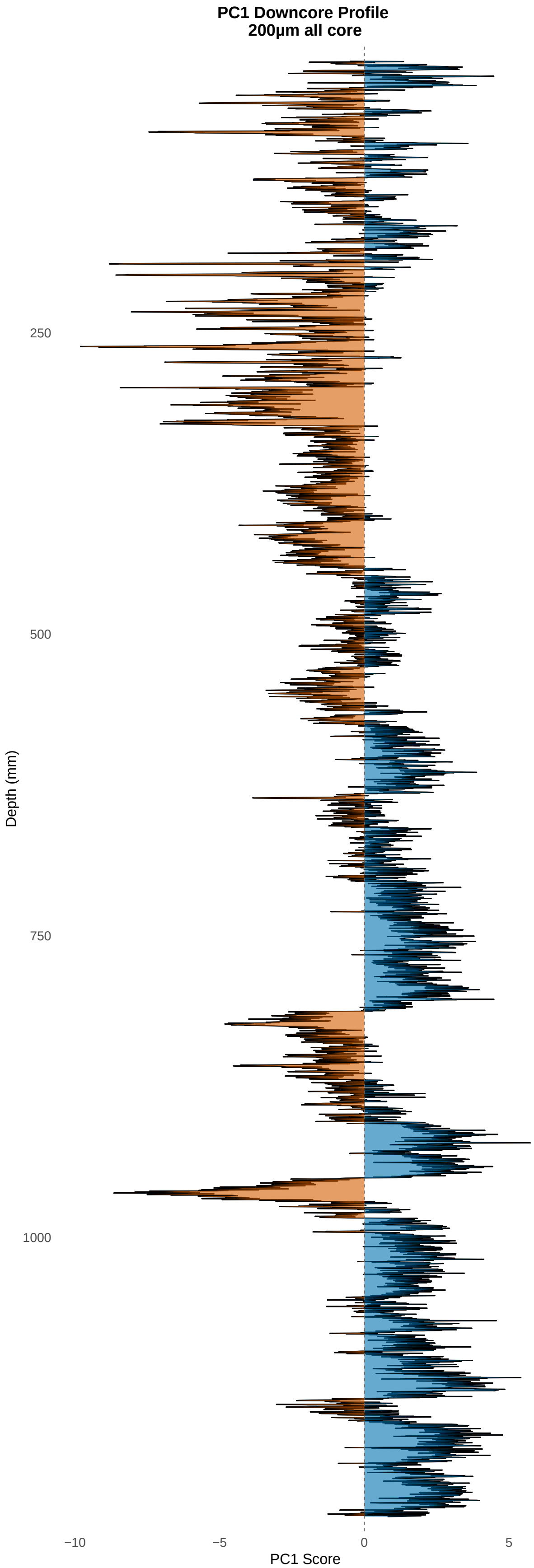


PC1 vs PC3 - K-means cluster facteting

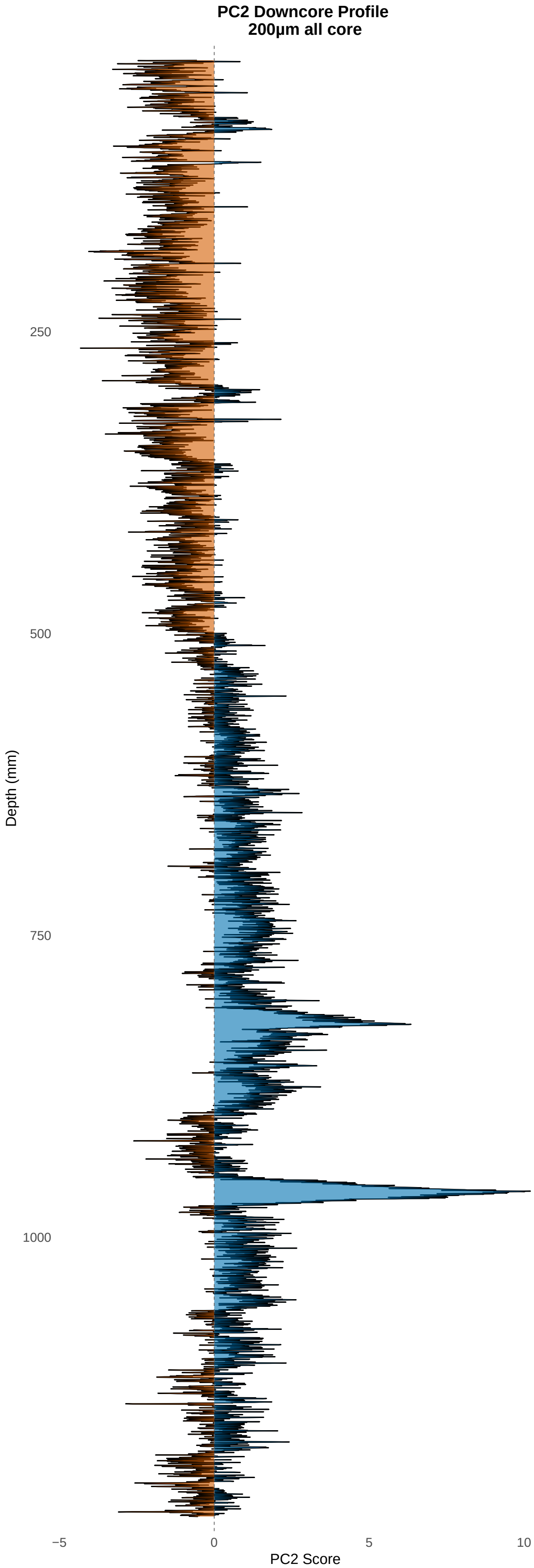


Downcore variations of Principal Components

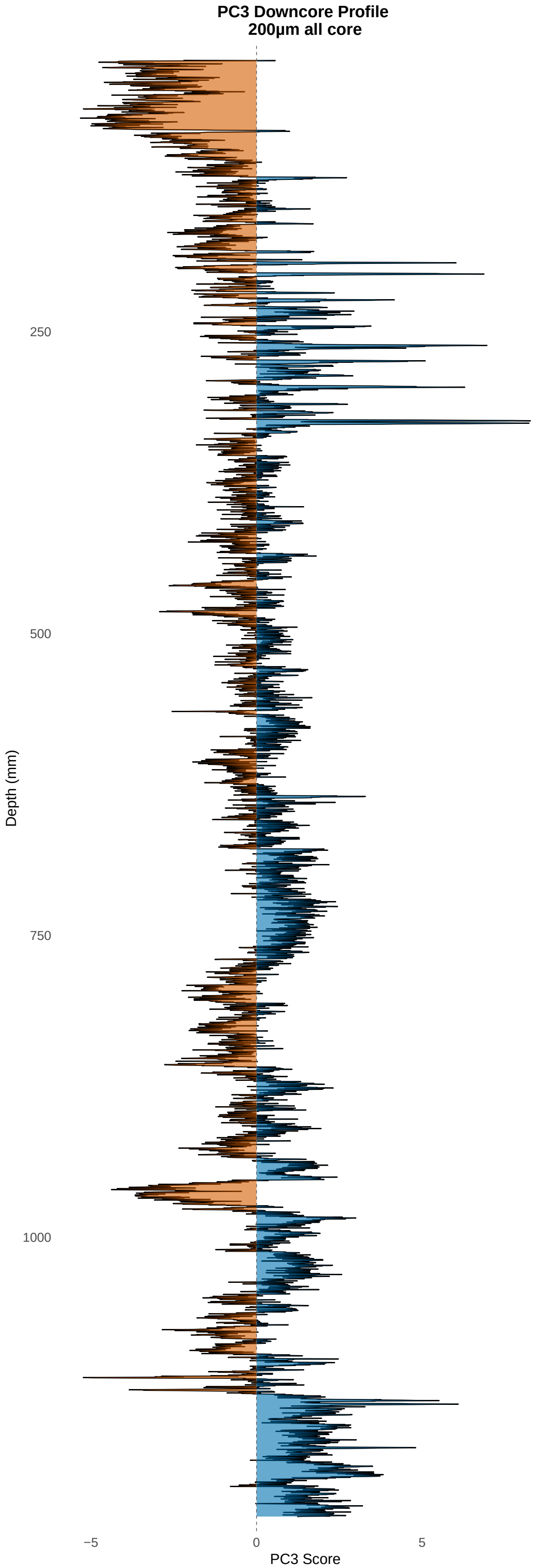
PC1



PC2


PC2 Downcore Profile
200µm all core
250
500
750
1000
Depth (mm)
−5
0
5
10
PC2 Score

PC3


PC3 Downcore Profile
200µm all core
250
500
750
1000
Depth (mm)
−5
0
5
PC3 Score